

Final Vulnerability Assessment Report

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Target IP: 10.11.0.1

1. Vulnerability Scan Results

Configuration: `nmap -sV --script vuln 10.11.0.1`

Primary Finding:

- **Vulnerability Name:** Avahi Denial of Service (DoS)
- **ID:** CVE-2011-1002
- **Status:** Detected in Pre-scan (Broadcast discovery)
- **Description:** The scan identified that the network configuration for Avahi (mDNS) is susceptible to a Denial of Service attack via a NULL UDP packet.

2. Vulnerability Classification Table

Port/Service	Finding	CVE ID	Classification	Risk Level
Network (UDP)	Avahi Packet DoS	CVE-2011-1002	Denial of Service	Medium
80 / 8080	HTTP Services	Cleartext Port	Configuration	Medium
6789	IBM-DB2-Admin	Exposed Admin	Access Control	High

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3. Methodology & Potential Security Implications

Methodology Explanation: I used the **Nmap Scripting Engine (NSE)** to perform a non-intrusive vulnerability sweep. The methodology involved "Pre-scanning," which identifies network-wide broadcast vulnerabilities before even checking specific ports. This allowed me to find the **CVE-2011-1002** vulnerability, which affects how the host handles specific network packets.

Security Implications: The discovery of **CVE-2011-1002** indicates that an attacker on the same local network could crash the Avahi daemon or degrade the performance of the gateway by sending a single, malformed UDP packet.

Furthermore, the discovery of **Port 6789 (IBM-DB2-Admin)** and **Port 53 (Domain)** on the same host suggests this is a "Multi-Role Server." If an attacker uses a DoS attack to crash the services on this machine, it would take down both the database administration and the network's name resolution (DNS), causing a total lab network outage.

4. Conclusion & Recommendation

- **Update Services:** Update the Avahi-daemon to a version later than 0.6.29 to patch CVE-2011-1002.
- **Filter Broadcasts:** Configure the firewall to drop malformed or NULL UDP packets at the network edge.
- **Restrict Admin Ports:** Port 6789 should be restricted to a specific management IP rather than being open to the whole lab subnet.